

Stony Brook Institute at Anhui University
2022 Fall Semester
***Fundamentals of Programming* Final Exam (Paper A)**
Reference Answers and Scoring Criteria

I. Multiple Choices (10 items, 1 point each)

No.	1	2	3	4	5	6	7	8	9	10
Answer	C	A	A	B	B	C	D	D	C	A

II. Blank Fillings (10 items, 2 points each)

- (1) `i < nums.length`
- (2) `max = nums[i]`
- (3) `x < width`
- (4) `line(x, 0, x, height)`
- (5) `translate(width/2, height/2)`
- (6) `text(message, 0, 0)`
- (7) `loadPixels()`
- (8) `x + y*width`
- (9) `movieEvent`
- (10) `image(movie, 0, 0)`

III. Corrections (5 items, 2 points each)

- 1. `for (int i = 0; i < numbers.length; i++)`
- 2. `size(200, 200, P3D);`
- 3. `PImage img = loadImage("flower.jpg");`
- 4. `Car(int _x, int _y, int _c)`
- 5. `Car[] cars = new Car[18];`

IV. Programming (5 items, 60 points in total)

- 1. (10 points)

```
void setup() {  
    size(400, 200);  
}
```

```
void draw() {
    background(255);
    branch(width/2, 0, 100);
}
```

```
void branch(float x, float y, float h) {
    line(x, y, x-h, y+h);
    line(x, y, x+h, y+h);
    if (h > 2) {
        branch(x-h, y+h, h/2);
        branch(x+h, y+h, h/2);
    }
}
```

2. (10 points)

```
int x = 0;
int speed = 1;
```

```
void setup() {
    size(640, 360);
}
```

```
void draw() {
    background(255);
    x = x + speed;
    if (x > width || x < 0) {
        speed = speed * -1;
    }
    stroke(0);
    fill(175);
    ellipse(x, 100, 32, 32);
}
```

3. (10 points)

```
PImage img;
float rot = 0.0;
```

```
void setup() {
```

```

    size(640, 360);
    img = loadImage("sunflower.jpg");
}

```

```

void draw() {
    background(255);
    translate(width/2, height/2);
    rotate(rot);
    imageMode(CENTER);
    image(img, 0, 0);
    rot += 0.01;
}

```

4. (15 points)

```

import processing.sound.*;

```

```

class Doorbell {
    float x, y;
    float r;
    SoundFile dingdong;

```

```

    Doorbell(float _x, float _y, float _r, String filename, PApplet t) {
        x = _x;
        y = _y;
        r = _r;
        dingdong = new SoundFile(t, filename);
    }

```

```

    void ring() {
        if (!dingdong.isPlaying())
            dingdong.play();
    }

```

```

    boolean contains(float mx, float my) {
        if (dist(mx, my, x, y) < r) {
            return true;
        } else {
            return false;
        }
    }

```

```

    }
}

void display(float mx, float my) {
    if (contains(mx, my)) {
        fill(100);
    } else {
        fill(175);
    }
    stroke(0);
    strokeWeight(4);
    ellipse(x, y, 2*r, 2*r);
}
}

```

5. (15 points)

```
Planet[] planets = new Planet[8];
```

```

void setup() {
    size(640, 640);
    for (int i = 0; i < planets.length; i++) {
        planets[i] = new Planet(64 + i*32, 24);
    }
}

```

```

void draw() {
    background(255);
    pushMatrix();
    translate(width/2, height/2);
    stroke(0);
    fill(255);
    ellipse(0, 0, 64, 64);
    for (int i = 0; i < planets.length; i++) {
        planets[i].update();
        planets[i].display();
    }
    popMatrix();
}

```

```

class Planet {
    float theta;
    float diameter;
    float distance;
    float orbitspeed;

    Planet(float _distance, float _diameter) {
        distance = _distance;
        diameter = _diameter;
        theta = 0;
        orbitspeed = random(0.01, 0.03);
    }

    void update() {
        theta += orbitspeed;
    }

    void display() {
        pushMatrix();
        rotate(theta);
        translate(distance, 0);
        stroke(0);
        fill(175);
        ellipse(0, 0, diameter, diameter);
        popMatrix();
    }
}

```